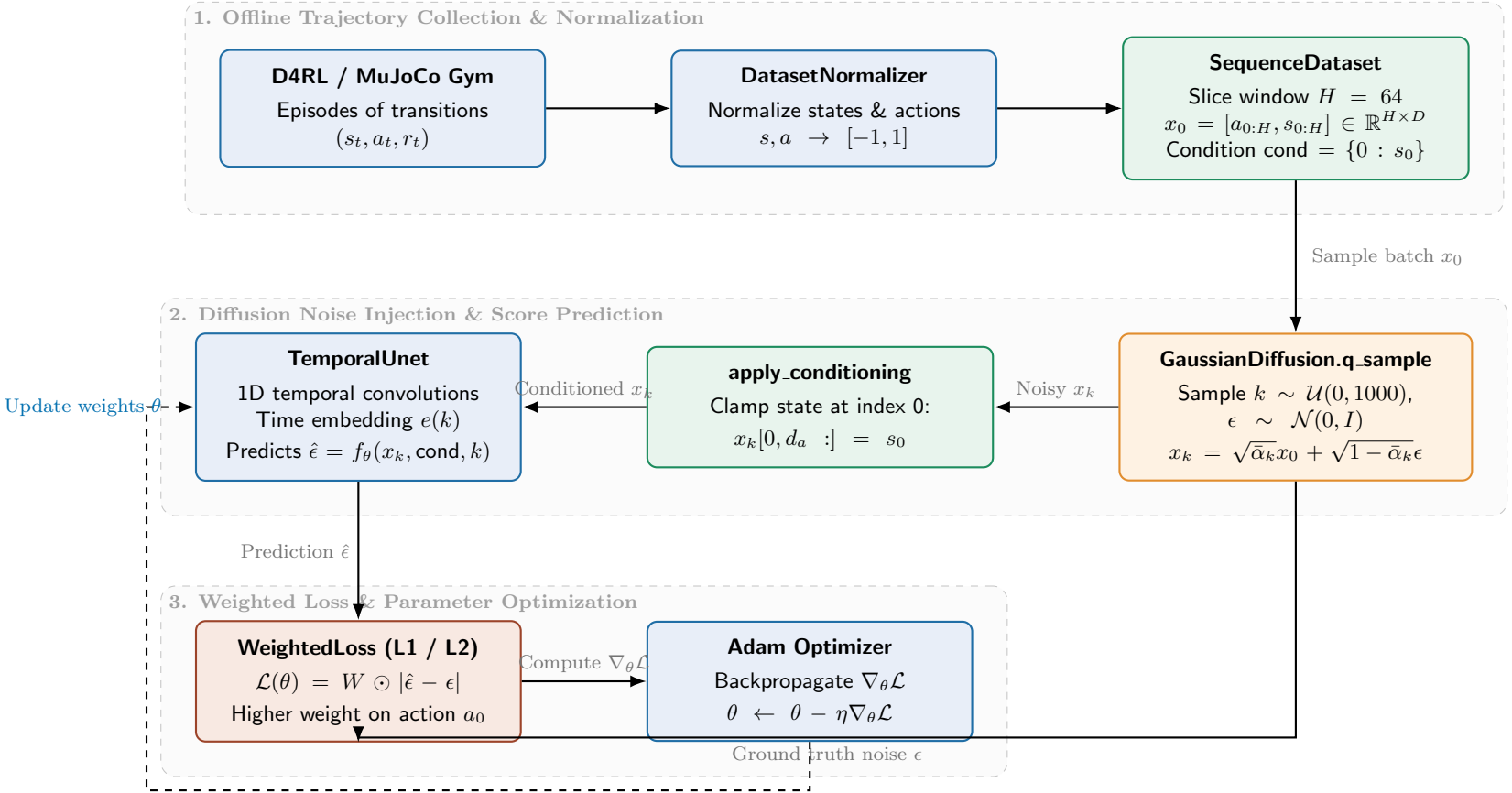

1. Offline Trajectory Collection & Normalization
D4RL / MuJoCo Gym
Episodes of transitions
(s_t, a_t, r_t)
DatasetNormalizer
Normalize states & actions
$s, a \rightarrow [-1, 1]$
SequenceDataset
Slice window $H = 64$
$x_0 = [a_{0:H}, s_{0:H}] \in \mathbb{R}^{H \times D}$
Condition cond $= \{0 : s_0\}$
Sample batch x_0
2. Diffusion Noise Injection & Score Prediction
TemporalUnet
1D temporal convolutions
Time embedding $e(k)$
Predicts $\hat{\epsilon} = f_\theta(x_k, \text{cond}, k)$
Conditioned x_k
apply_conditioning
Clamp state at index 0:
$x_k[0, d_a :] = s_0$
Noisy x_k
GaussianDiffusion.q_sample
Sample $k \sim \mathcal{U}(0, 1000)$,
$\epsilon \sim \mathcal{N}(0, I)$
$x_k = \sqrt{\bar{\alpha}_k} x_0 + \sqrt{1 - \bar{\alpha}_k} \epsilon$
Update weights θ
Prediction $\hat{\epsilon}$
3. Weighted Loss & Parameter Optimization
WeightedLoss (L1 / L2)
$\mathcal{L}(\theta) = W \odot |\hat{\epsilon} - \epsilon|$
Higher weight on action a_0
Compute $\nabla_\theta \mathcal{L}$
Adam Optimizer
Backpropagate $\nabla_\theta \mathcal{L}$
$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
Ground truth noise ϵ